

COMPANION
to

ChatGPT Teaches TikZ

Chapter 9: Better Paths and Curves (I)

Contents

9.1	What This Companion Adds	2
9.2	A Path Vocabulary	3
9.3	Cubic Bézier Curves	3
9.4	Directional Curves with <code>to</code>	5
9.5	Closing a Path	6
9.6	Connecting Nodes with <code>edge</code>	7
9.7	Choosing the Simplest Construction	7
9.8	Worked Project: Curves in a Triangle	8
9.9	Debugging Workshop	10
9.10	Exercises	11
9.11	Solutions	11
9.12	ChatGPT Workshop	13
9.13	Final Checklist	13

9.1 What This Companion Adds

Chapter 9 introduced controlled curves, closed paths, and the `edge` operation. This Companion develops the judgment needed to choose among precise Bézier control points, directional curves, straight segments, and node edges.

The objective is not to make every path elaborate. It is to select the shortest description that preserves the shape or relationship that matters.

Design

Choose a path command by meaning. Use complete control only when the geometry requires it; otherwise prefer the simpler construction.

9.2 A Path Vocabulary

Form or option	Purpose
--	Draws a straight segment.
to	Draws a flexible connection.
controls P and Q	Supplies two Bézier control points.
out=45	Sets the direction leaving the start.
in=135	Sets the direction approaching the end.
looseness=1.4	Adjusts the breadth of a directional curve.
cycle	Closes the current path.
edge	Creates a connection from a named node.

9.3 Cubic Bézier Curves

A cubic Bézier curve has two endpoints and two control points.

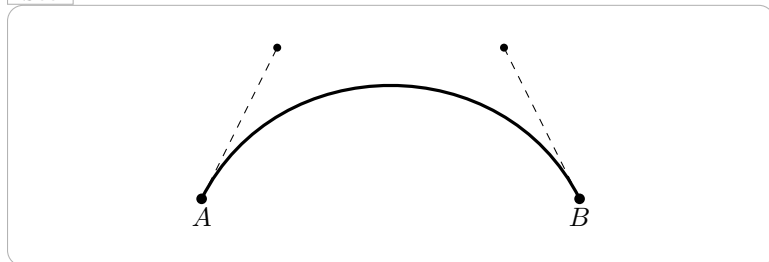
```
\draw (A)
  .. controls (P) and (Q) ..
  (B);
```

The curve begins at A in the direction of P and approaches B from the direction of Q. It does not normally pass through either control point.

```
\begin{tikzpicture}
  \coordinate (A) at (0,0);
  \coordinate (P) at (1,2);
  \coordinate (Q) at (4,2);
  \coordinate (B) at (5,0);

  \draw[dashed] (A) -- (P) (Q) -- (B);
  \draw[very thick]
    (A) .. controls (P) and (Q) .. (B);
\end{tikzpicture}
```

See



The dashed control lines make the starting and ending directions visible.

9.3.1 Changing One Control Point

Moving P changes how the curve leaves A; moving Q changes how it approaches B. Edit one point at a time so that the effect remains easy to diagnose.

Common Mistake

A cubic Bézier curve requires two control points separated by **and**. The form `controls (P)` is incomplete.

9.4 Directional Curves with `to`

When only the departure and arrival directions matter, use `out` and `in`.

```
\draw (A) to[out=45,in=135] (B);
```

Angles are measured in degrees. The curve leaves A at 45 degrees and approaches B from 135 degrees.

```
\begin{tikzpicture}
  \coordinate (A) at (0,0);
  \coordinate (B) at (5,0);
  \draw[very thick]
    (A) to[out=45,in=135] (B);
\end{tikzpicture}
```

See



This form is usually easier to revise than two explicit control points.

9.4.1 Adjusting looseness

Keep the endpoints and directions fixed while comparing curvature:

```
\draw (A)
  to[out=45,in=135,looseness=.7] (B);
\draw[very thick] (A)
  to[out=45,in=135,looseness=1.5] (B);
```

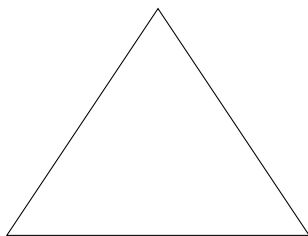
A larger value produces a broader curve. Change `looseness` only after the `out` and `in` directions are sensible.

9.5 Closing a Path

Use `cycle` whenever a path returns to its starting point.

```
\draw (0,0) -- (4,0) -- (2,3) -- cycle;
```

See



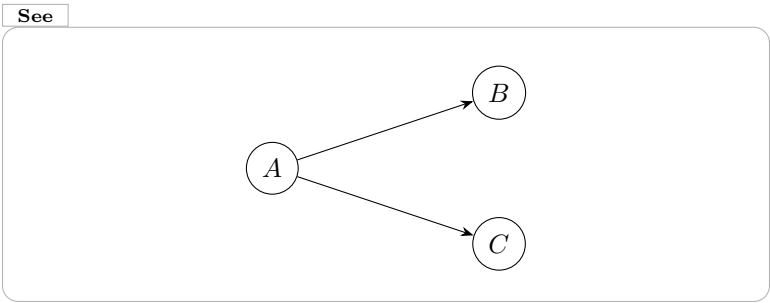
Writing the starting coordinate again can produce the same picture, but `cycle` states the intention directly and avoids repeating a coordinate.

9.6 Connecting Nodes with edge

The `edge` operation is useful when one named node has several connections.

```
\begin{tikzpicture}
  \node[draw,circle] (A) at (0,0) {$A$};
  \node[draw,circle] (B) at (3,1) {$B$};
  \node[draw,circle] (C) at (3,-1) {$C$};

  \path (A)
    edge[-{Stealth}] (B)
    edge[-{Stealth}] (C);
\end{tikzpicture}
```



The source groups two relationships that share the same starting node.

9.7 Choosing the Simplest Construction

Use	When
--	The connection should remain straight.
to[out,in]	Entry and exit directions are sufficient.
Bézier controls	The exact character of the curve matters.

Use	When
edge	Named nodes participate in a connection network.
cycle	The current path must close.

Do not choose a more detailed command merely because it is available.

9.8 Worked Project: Curves in a Triangle

The project uses one closed path, one controlled Bézier curve, and two directional curves.

9.8.1 Stage 1: Define the Vertices

```
\coordinate (A) at (0,0);
\coordinate (B) at (5,0);
\coordinate (C) at (2.5,3);
```

9.8.2 Stage 2: Draw the Closed Path

```
\draw[thick] (A) -- (B) -- (C) -- cycle;
```


9.8.3 Stage 3: Add the Curves

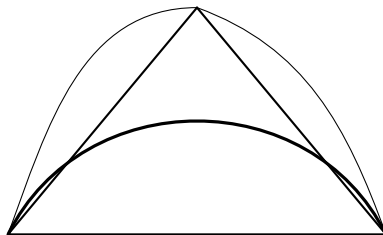
```
\draw[very thick] (A)
  .. controls (1,2) and (4,2) .. (B);
\draw (A) to[out=70,in=180] (C);
\draw (C) to[out=-20,in=110] (B);
```

9.8.4 The Complete Picture

```
\begin{tikzpicture}
  \coordinate (A) at (0,0);
  \coordinate (B) at (5,0);
  \coordinate (C) at (2.5,3);

  \draw[thick] (A) -- (B) -- (C) -- cycle;
  \draw[very thick] (A)
    .. controls (1,2) and (4,2) .. (B);
  \draw (A) to[out=70,in=180] (C);
  \draw (C) to[out=-20,in=110] (B);
\end{tikzpicture}
```

See



Design

Each path has a reason: straight segments define the triangle, explicit controls define the principal curve, and directional options define the two simpler curves.

9.9 Debugging Workshop

9.9.1 Using Only One Control Point

Incorrect:

```
\draw (A) .. controls (P) .. (B);
```

Correct:

```
\draw (A) .. controls (P) and (Q) .. (B);
```

9.9.2 Putting Curve Options on --

Use `to` when a connection needs `out`, `in`, or `looseness`.

```
\draw (A) to[out=30,in=150] (B);
```

9.9.3 Choosing Inconsistent Directions

If a curve loops unexpectedly, inspect `out` and `in` before changing `looseness`. The directions establish the basic shape.

9.9.4 Forgetting `cycle`

An open path cannot enclose or fill a region. Add `-- cycle` when the geometry is meant to close.

9.10 Exercises

1. Draw a cubic Bézier curve and show its control lines.
2. Move only the first control point and describe the change.
3. Draw the same endpoints with `to[out,in]`.
4. Compare two values of `looseness`.
5. Draw a closed quadrilateral with `cycle`.
6. Connect one node to three others with `edge`.
7. Draw two parallel curved arrows by using opposite bends.
8. Revise the worked project by changing only curve parameters.

9.11 Solutions

9.11.1 Exercises 1–4

```
\draw[dashed] (A) -- (P) (Q) -- (B);  
\draw (A) .. controls (P) and (Q) .. (B);
```

For Exercise 2, change only the coordinate assigned to P; the curve's approach to B should remain governed by Q.

```
\draw (A) to[out=45,in=135] (B);
```

```
\draw (A) to[out=45,in=135,looseness=.7] (B);  
\draw (A) to[out=45,in=135,looseness=1.4] (B);
```

9.11.2 Exercises 5–7

```
\draw (A) -- (B) -- (C) -- (D) -- cycle;
```

```
\path (A)  
  edge (B)  
  edge (C)  
  edge (D);
```

```
\draw[->] (A) to[bend left=25] (B);  
\draw[->] (B) to[bend left=25] (A);
```

For Exercise 8, preserve A, B, and C. Edit only the control points, angles, or `looseness` values.

9.12 ChatGPT Workshop

Ask ChatGPT to explain the geometric effect of every revision rather than merely producing a different-looking curve.

1. Compare `--`, `to`, `edge`, Bézier controls, and `cycle` in this diagram.
2. Rewrite this curve with `out` and `in` while preserving its endpoints and approximate shape.
3. Show the control lines for every Bézier curve in this picture.
4. Diagnose this unexpected loop by checking `out`, `in`, and `looseness` in that order.
5. Simplify this path by replacing unnecessary control points with a directional `to` curve.

9.13 Final Checklist

Before continuing to Chapter 10, check that you can

1. construct a cubic Bézier curve with two control points;
2. explain how each control point affects an endpoint;
3. draw a directional curve with `out` and `in`;
4. adjust a curve with `looseness`;
5. close a path with `cycle`;
6. connect one named node to several others with `edge`; and
7. choose the simplest path construction that expresses the geometry.